Medicare+

Problem Statement: Medication Adherence

Domain: Healthcare

Team : Green Pixel

Member 01 : Priyadharshini A

Impact: Poor health outcomes, preventable hospital visits, and medication errors.

The Problem

Busy schedules & frequent forgetfulness lead to missed or delayed daily medication doses.

Complex handwritten prescriptions make it hard for patients to understand correct dosages.

Existing reminder apps only send basic notifications and lack intelligent support.

Who is affected?

Chronic patients → Elderly individuals → Family caregivers

Impact: Poor health outcomes, preventable hospital visits, and medication errors

WHY THIS PROBLEM EXISTS

Fast-paced modern lifestyles lead to chronic forgetfulness and missed doses.

Prescriptions feature messy handwriting ,complex medical terminology that confuse patients.

Traditional reminder apps only notify without tracking adherence or alerting caregivers.

V '

CORE PROBLEM -> Medication non-adherence.

USER CONSEQUENCE-> Worsening health conditions and health risks.

REAL IMPACT Preventable hospital visits and increased medical complications.

Current Approach:

Current Method → Friction → Delay / Cost / Risk → Poor Outcome

OUR SOLUTION

INTRODUCING **Medicare+** enables patients to manage medications seamlessly through smart reminders, automated prescription scanning, and AI guidance.

Patient or Caregiver
↓
Uploads a photo of a handwritten medical prescription image
↓
Medicare+ (Offline Smart Prescription Scanner)
↓
Client-side Tesseract OCR text extraction & local medical database matching
↓
Structured drug breakdown (uses, prescribed dosages, and safety precautions) with smart reminders
↓
Error-free adherence, complete data privacy, and zero reliance on external APIs or internet connectivity

Core capabilities

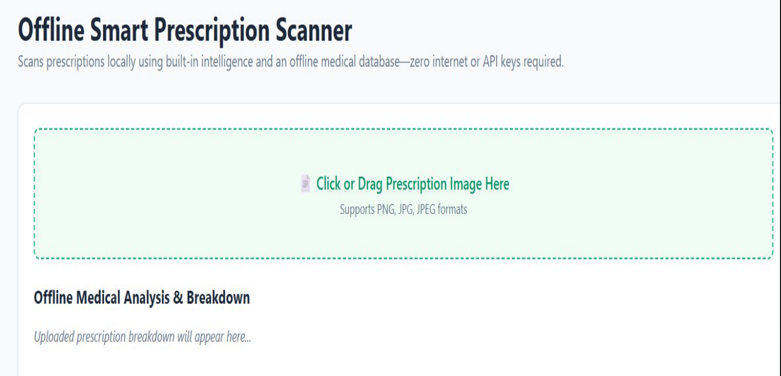
01. **OCR Prescription Scanner:** Instantly extracts and decodes medicine names and dosages from prescription images.

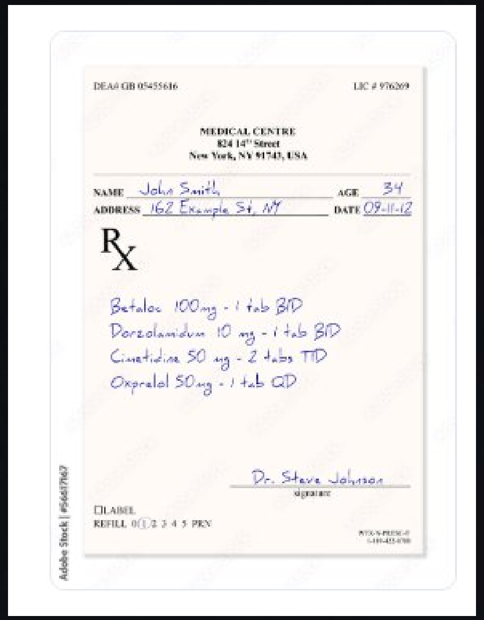
02 . **Smart Reminders & Alerts:** Keeps patients on track and notifies caregivers of missed doses.

03 . **AI Medical Assistant:** Provides instant, clear answers regarding medication usage and safety guidelines.

PROTOTYPE / PRODUCT EXPERIENCE

Prototype Screenshots





Right: User journey

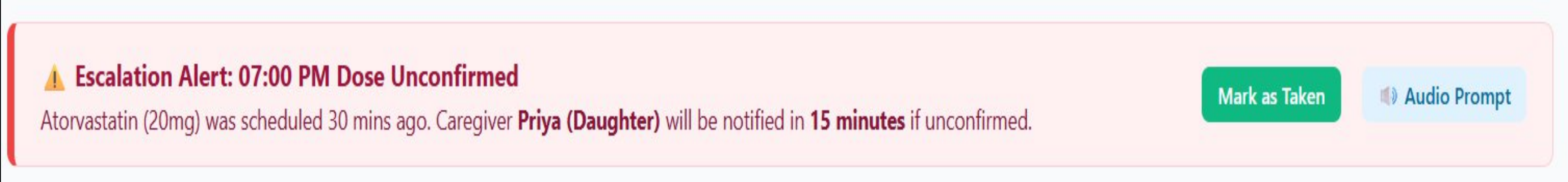
01 — **INPUT:** User uploads or snaps a photo of their handwritten prescription.

02 — **PROCESS:** Tesseract OCR extracts text locally while the app processes the medical data.

03 — **OUTPUT:** A clear breakdown of medicine names, uses, dosages, and safety precautions appears on screen.

04 — **ACTION:** The app automatically schedules smart reminders and logs the medication.

Prototype Screenshots

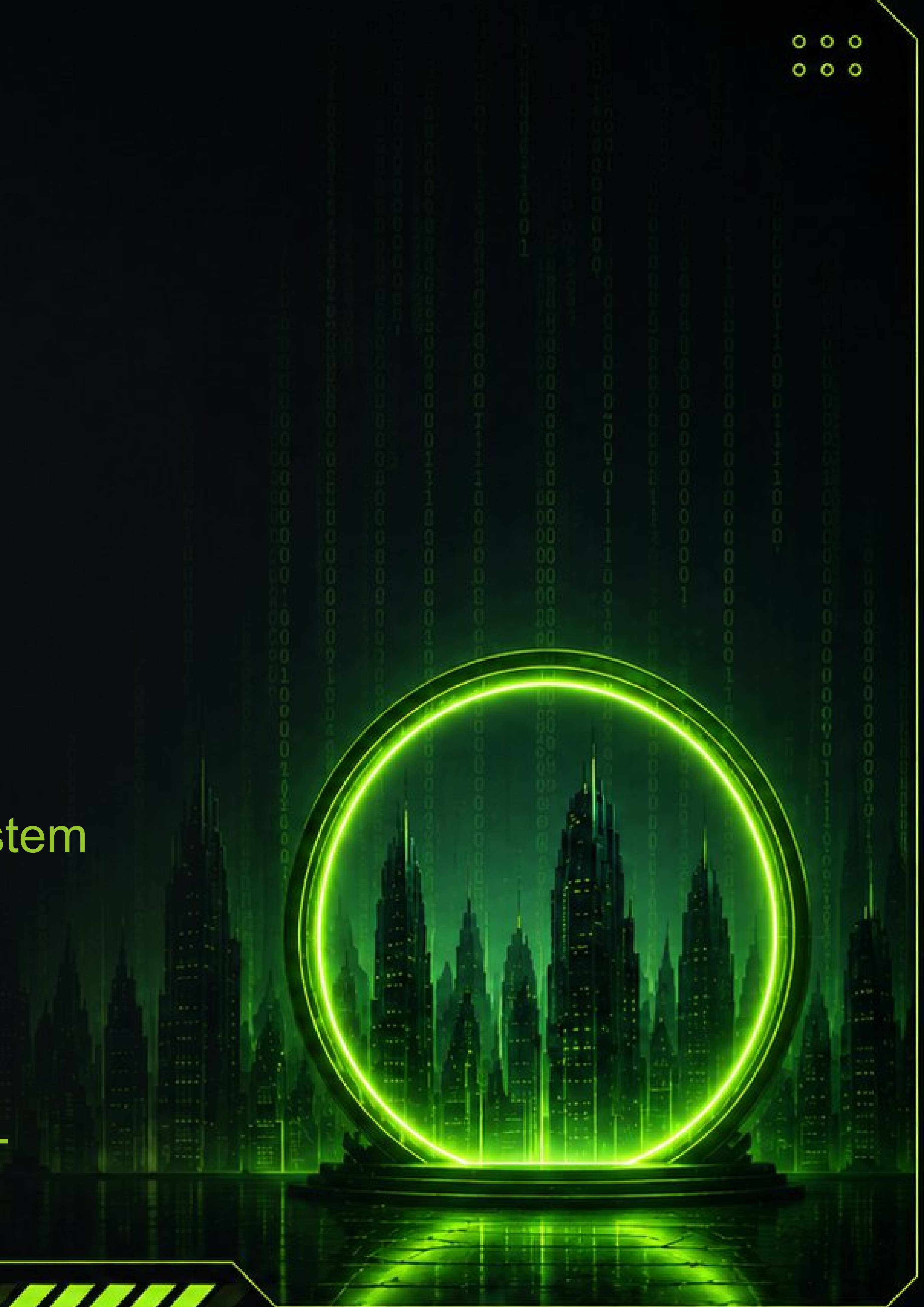


The audio plays here

WHAT MAKES US DIFFERENT?

WHY THIS APPROACH?

Existing Approach	Medicare+
Manual entry / basic alarms	Automated OCR prescription scanning
No caregiver integration	Built-in caregiver alert system
Generic notifications guidance	Personalized AI medical & breakdown
Requires complex setup capable	Fast, intuitive, and offline- client-side processing

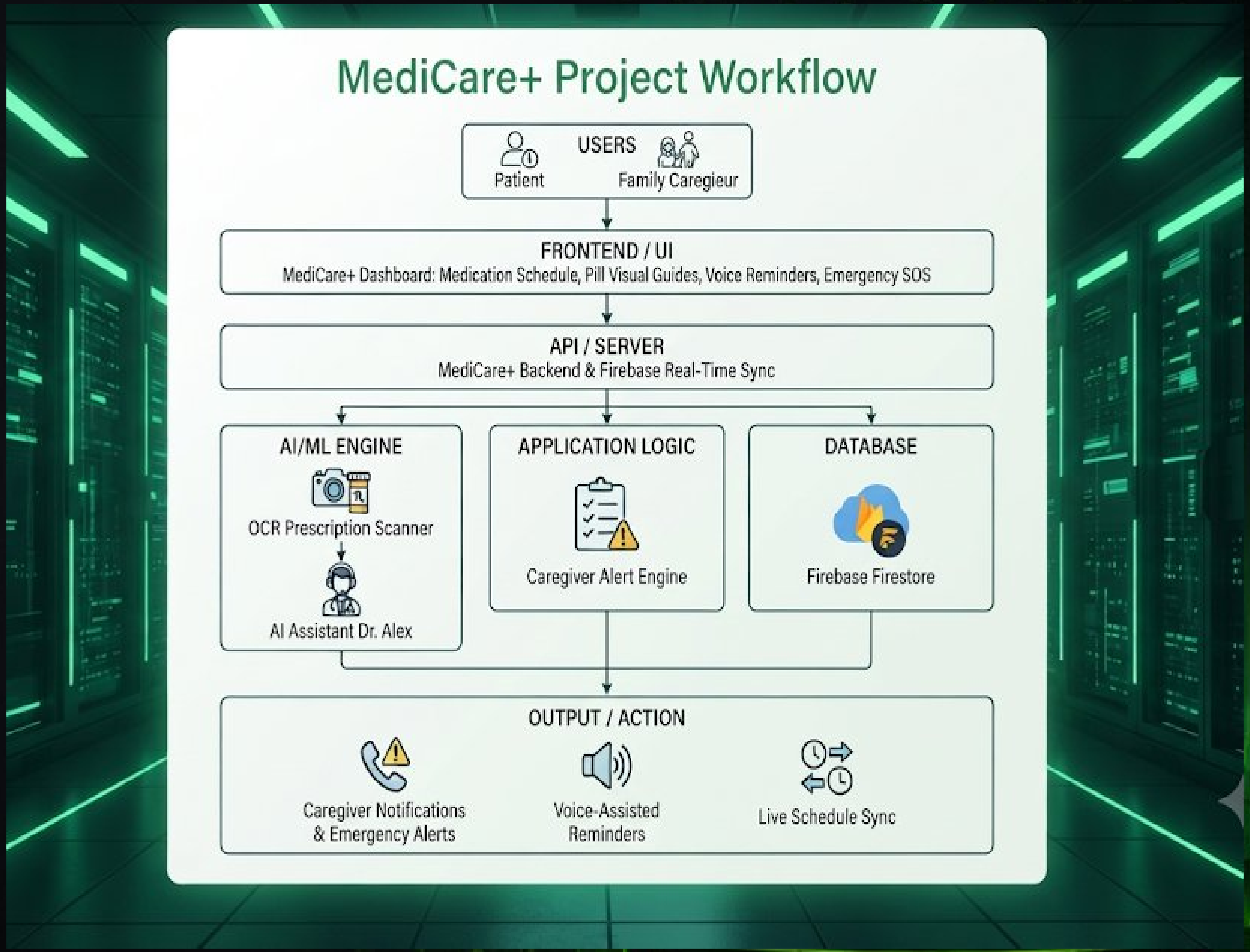


TECHNICAL ARCHITECTURE

Tech stack:

Frontend
HTML, CSS, Javascript
Backend
Firebase Functions
Database
Firebase Firestore
AI/ML
OCR Prescription Scanner & Dr. Alex AI Health Assistant
Deployment
Firebase Hosting

Example Workflow Image



FEASIBILITY + IMPACT

FEASIBILITY

TECHNICAL

- Modern web/mobile frameworks, cloud databases, and established OCR and AI model APIs.
- Firebase Firestore for real-time medication scheduling and status updates, OCR prescription scanner, and the Dr. Alex AI health assistant.
- Modular component development linking real-time database syncing with automated notifications, pill visual guides, and voice-assisted reminders.

OPERATION

- Cloud infrastructure hosting, development workspace, and testing devices for patient and caregiver interfaces.
- Cloud-backed application with secure, real-time data handling.

HACKATHON MVP

- An interactive dashboard integrating the OCR prescription scanner, Dr. Alex AI assistant, automated caregiver alert system, pill visual guides, and emergency SOS functionality.

IMPACT

Who benefits?

- Patients managing complex medication routines (such as elderly or chronically ill individuals) and their family caregivers.

What improves?

- Overall medication adherence rates, reduction of missed doses, and minimized emergency health risks.

Why does it matter?

- Prevents severe medical complications stemming from non-adherence and provides immediate safety nets via emergency alerts and caregiver notifications.

IMPLEMENTATION ROADMAP

WHAT HAPPENS NEXT?

NOW

Core workflow & Functional UI

OCR prescription scanner integration

Firebase Firestore real-time database setup

→

NEXT

Dr. Alex AI health assistant refinement

Automated caregiver alerts for missed doses

Initial user testing and reliability checks

→

LATER

Real-world pilot with patients or caregivers

Emergency SOS integration and voice-assisted reminders

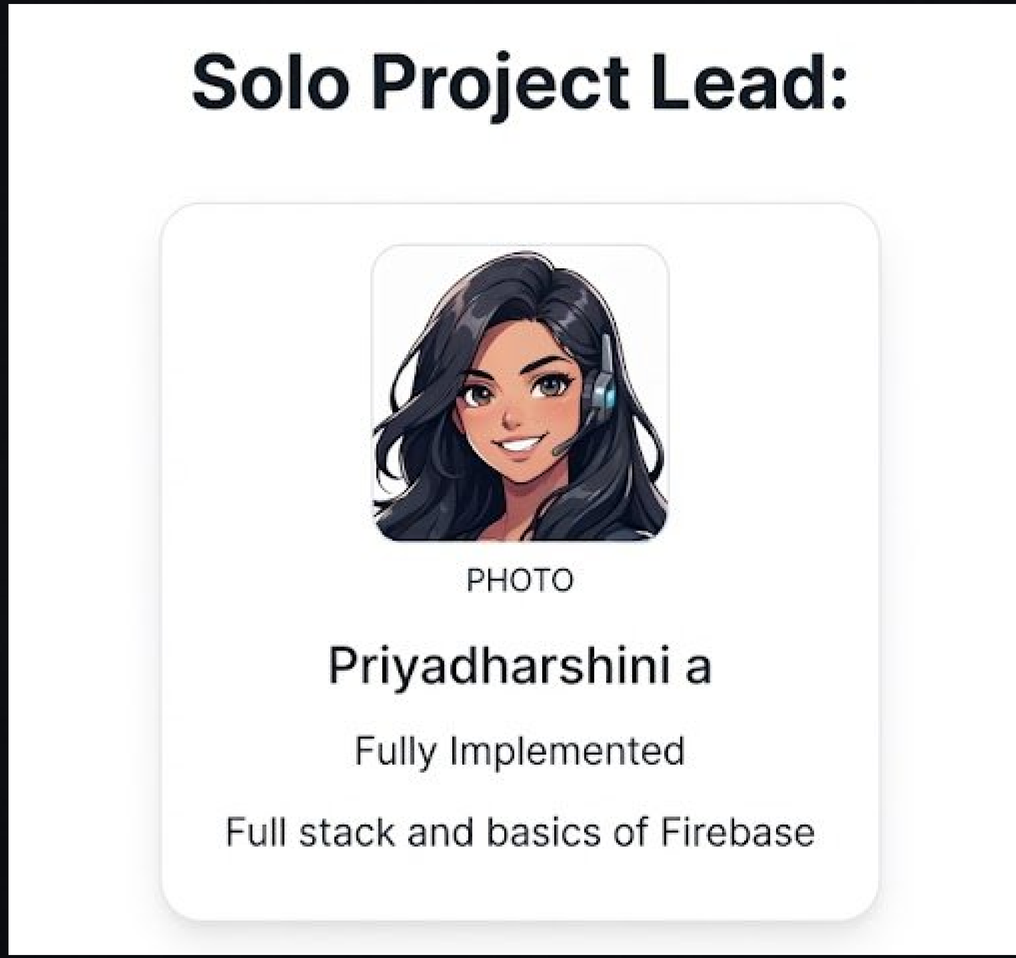
→

SCALE

Expanding Medication database domains and multi-user support

TEAM + EXECUTION

Example Image



I possess the full-stack development expertise and AI/ML integration skills required to rapidly deliver and scale the Medicare+ solution.